European Cross-Commodity Risk Monitor

Gas Tightness · Carbon Supply Signal · Power Curve Implications

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Signal Summary

■ Bullish	■ Neutral	■ Bearish
3	2	3

Market Narrative

EU gas storage remains critically below seasonal norms, creating an acute injection problem that the market has yet to fully resolve. The TTF prompt reflects ongoing Hormuz risk and Qatari supply uncertainty; any further escalation in a thin summer market would be felt immediately across the curve.

Carbon is holding in the mid-70s, contributing roughly €29/MWh to gas plant variable costs. The implied EUA in Cal+1 power is well below spot, suggesting the forward power curve is either pricing EUA weakness or significant renewable displacement — both of which look optimistic against current structural supply signals.

German Cal+1 looks directionally cheap relative to the gas and carbon fundamental stack. The CSS is negative, but this reflects structural renewable compression of baseload rather than true cheapness. The primary upside trigger is a cold autumn onset with storage still below 75%; the primary downside is a Hormuz resolution compressing TTF by €8–12/MWh.

1 | Gas Tightness

TTF M+1	**€49.00/MWh** (+33.8% YoY)
EU Storage	**36.3%** (+5.8pp vs 5-yr avg)
Required injection	**3682 GWh/day** | Current: 2100 GWh/day | Gap: +1582
Norwegian supply	Plateau ~114.9 Bcm 2025; 2026 guide 110–120 Bcm
LNG shock	Qatar force majeure ~17% of global liquefaction (March 2026)

Chart 1 — EU Natural Gas Storage vs Seasonal Norms | Source: GIE AGSI+

Chart 1: EU gas storage fill % vs 5-year seasonal band. Current: 36.3% (+5.8pp vs avg). Source: GIE AGSI+.

2 | Carbon Signal

EUA front-year	**€73.85/t**
Carbon cost per MWh	**€29.10/MWh** (EUA × 0.394 tCO■/MWh)
MSR withdrawal	275.5 Mt removed Sep 2025–Aug 2026 | Supply tightens Sep 2026
ETS Directive review	July 2026 | Key policy catalyst

3 | Power Curve Implications

Clean Spark Spread:

CSS = Power − (Gas ÷ η) − (EF × Carbon) = 100.27 − 99.74 − 29.10 = **€-28.56/MWh**

Implied EUA (solving CSS = 0):

EUA* = (Power − Gas/η) ÷ EF = **€1.35/t** vs spot €73.85/t → gap: €72.50/t

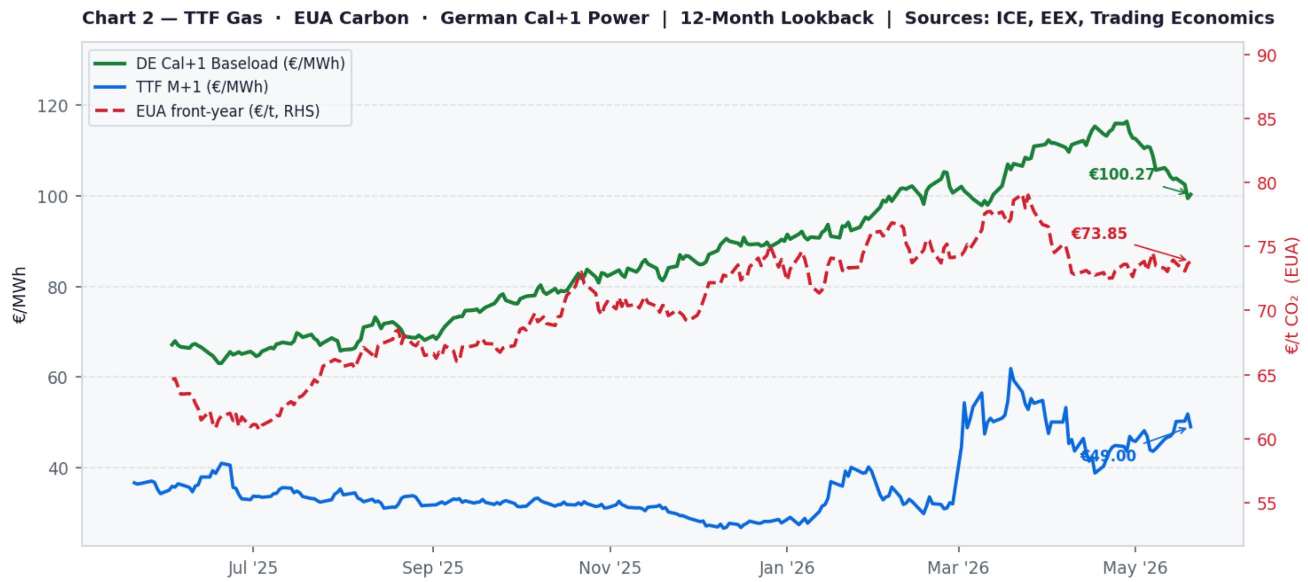

Chart 2: TTF M+1 (blue), EUA front-year (red, RHS), German Cal+1 baseload (green). 12-month lookback. Sources: ICE, EEX, Trading Economics.

4 | Monitor Metrics Dashboard

Metric	Value	Signal	Relevance
TTF M+1 (€/MWh)	€49.00 (+33.8% YoY)	▲ BULLISH	Primary power price input
EU Storage vs 5-yr avg	36.3% (+5.8pp)	▼ BEARISH	Low buffer amplifies demand shocks
Required injection pace	3682 GWh/d	▼ BEARISH	Gap vs target = winter risk
EUA front-year (€/t)	€73.85	→ NEUTRAL	+€30/MWh carbon cost in power
Carbon cost per MWh	€29.10/MWh	→ NEUTRAL	Direct variable cost floor
Clean Spark Spread	€-28.56/MWh	▼ BEARISH	Gas plant profitability
German Cal+1 baseload	€100.27/MWh	▲ BULLISH	Forward power curve anchor
Implied EUA in Cal+1	€1.35/t (gap: €72.50)	▲ BULLISH	Carbon unpriced in forward power

5 | Risk Skew

Upside catalysts (bullish power):

- ▲ Cold autumn onset — every 1pp below seasonal norm at 1 Nov adds ~€3–5/MWh to front-winter power.
- ▲ Hormuz re-escalation — TTF could retest €60+; Cal+1 power follows within 24–48h.
- ▲ Hawkish ETS Directive review (July 2026) — EUA above €90/t adds ~€6/MWh to power variable cost.
- ▲ Norwegian unplanned outage — 10 bcm/day flow cut moves TTF 3–5% intraday.

Downside catalysts (bearish power):

- ▼ Hormuz resolution / US-Iran deal — TTF -€8–12/MWh; Cal+1 power follows 1:1.
- ▼ Warm summer and autumn — 2°C above-average Sep/Oct adds 5–8pp to storage.
- ▼ US LNG wave on schedule — 2027 supply easing weighs on Cal+2 and beyond.
- ▼ EUA weakness — sustained below €65/t reduces carbon floor in power by €4–6/MWh.

Data sources: GIE AGSI+, ICE/Yahoo Finance (TTF, EUA), EEX/Yahoo Finance (German power). Generated: 2026-05-20 22:21. Not investment advice.